Diwali Sales Analysis Using Python

Project Title

Diwali Sales Analysis and Customer Purchase Behavior Insights Using Python

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1. Project Overview

The Diwali Sales Analysis project aims to explore customer purchasing patterns during the festive season and generate actionable business insights from historical sales data. The analysis was performed using Python in Jupyter Notebook by applying data cleaning, exploratory data analysis (EDA), and data visualization techniques.

The project focuses on identifying key customer segments, understanding purchasing behaviour across different demographics, and helping businesses make data-driven marketing and sales decisions.

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2. Problem Statement

Businesses generate large amounts of transactional data during festive seasons. However, without proper analysis, it becomes difficult to identify:

- Which customer segments contribute the most revenue.
- Which products are highly demanded.
- Which states and occupations generate maximum sales.
- The target customers for future marketing campaigns.

The objective of this project is to transform raw sales data into meaningful insights that support strategic decision-making.

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3. Objectives

The main objectives of this project are:

- Perform data cleaning and preprocessing.
- Analyse customer demographics and purchasing behavior.
- Identify top-performing states and product categories.
- Determine high-value customer segments.
- Generate insights for business growth and targeted marketing.

4. Tools and Technologies Used

Tool/Technology	Purpose
Python	Data Analysis and Programming
Jupyter Notebook	Development Environment
Pandas	Data Cleaning and Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization

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5. Dataset Information

The dataset contains customer transaction records during the Diwali festival season, including:

- User ID
- Gender
- Age Group
- Marital Status
- State
- Occupation
- Product Category
- Product ID
- Number of Orders
- Purchase Amount

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6. Data Preprocessing

The following preprocessing steps were performed:

Data Cleaning

- Imported CSV data using Pandas.